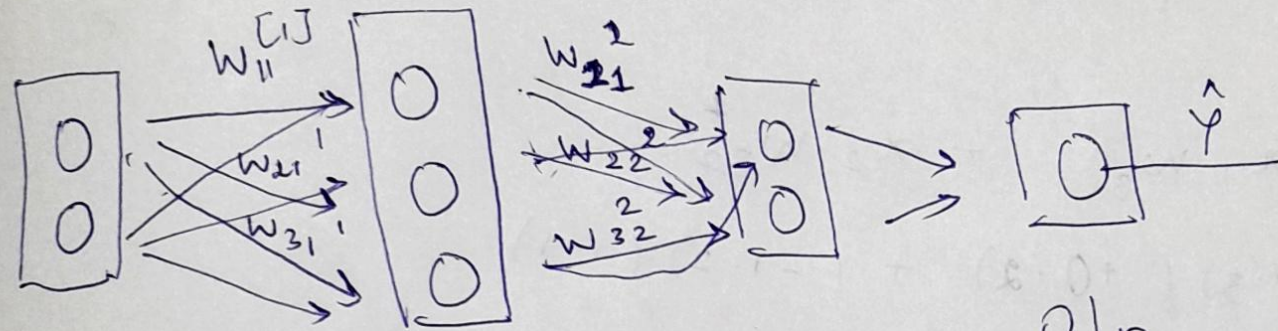


21 July 2026.

Numerical.

Fully Connected Network.



I/p

HL1

HL2

O/p

$$I/p \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0.3 \\ -1.2 \end{bmatrix}$$

$$W_{11}^1 = 0.1$$

$$W_{21}^1 = -0.1$$

$$W_{31}^1 = 0.2$$

$$W_{12}^1 = -1.1$$

$$W_{22}^1 = 0.4$$

$$W_{32}^1 = 1.1$$

$$W_{11}^2 = 0.2$$

$$W_{22}^2 = 0.3$$

$$W_{12}^2 = 0.1$$

$$W_{22}^2 = -0.1$$

$$W_{13}^2 = -0.2$$

$$W_{23}^2 = -0.1$$

$$W_{11}^3 = 0.3$$

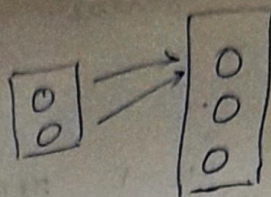
$$W_{12}^3 = -0.3$$

$$b = 0$$

$$g(z) = \frac{1}{1 + e^{-z}}$$

Sigmoid function

$\hat{y}$  : To find



HL1

$$z_1' = x_1 w_{11}' + x_2 w_{12}'$$

$$= (0.3)0.1 + (-1.2)(-1.1)$$

$$= 0.03 + 1.32 = 1.35$$

$$a_1' = g(z_1')$$

$$= \frac{1}{1 + 0.259}$$

$$= \frac{1}{1.259}$$

$$= \boxed{0.794}$$

$$z_2' = x_1 w_{21}' + x_2 w_{22}'$$

$$= (0.3)(-0.1) + (-1.2)(0.4)$$

$$= -0.03 + (-0.48)$$

$$= -0.51$$

$$a_2' = g(z_2')$$

$$= \frac{1}{1 + 1.665}$$

$$= \frac{1}{2.665}$$

$$= \boxed{0.375}$$

$$z_3' = x_1 w_{31}' + x_2 w_{32}'$$

$$= (0.3)(+0.2) + (-1.2)(1.1)$$

$$= +0.06 - 1.32$$

$$= -1.26$$

$$a_3' = g(z_3')$$

$$= \frac{1}{1 + 3.52}$$

$$= \frac{1}{4.52}$$

$$= \cancel{0.221}$$

$$= 0.221$$

112

$$Z_1^2 = a_1' W_{21}^2 + a_2' W_{12}^2 + a_3' W_{13}^2$$

$$= (0.794)(0.2) + (0.375)(0.1) + (0.221)(-0.23)$$

$$= 0.158 + 0.0375 - 0.0442$$

$$= 0.1513$$

$\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

 $\rightarrow$ 

$\begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$$a_1^2 = g(0.1513) = \frac{1}{1 + 0.8595} = \frac{1}{1.859}$$

$$= \boxed{0.537}$$

$$Z_2^2 = a_1' W_{21}^2 + a_2' W_{22}^2 + a_3' W_{23}^2$$

$$= 0.794(0.3) + 0.375(-0.1) + 0.221(-0.1)$$

$$= 0.238 + (-0.0375) - 0.0221$$

$$= 0.1784$$

$$a_2^2 = g(0.1784) = \frac{1}{1 + 0.836} = \frac{1}{1.836} = 0.544$$

O/p

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} > \begin{bmatrix} 0 \end{bmatrix}$$

$$Z_3^1 = 0.537(0.3) + (0.544)(-0.3)$$

$$= 0.1611 - 0.1632 = -0.0021$$

$$a_1^3 = \frac{1}{1 + 1.002} = \frac{1}{2.0021} = 0.49947$$

$$\hat{y} = 0.499 \approx 0.5$$